Distributed Model Predictive Control for Vehicle Platoon With Mixed Disturbances and Model Uncertainties

Xiaorong Hu, Lantao Xie, Lei Xie, Shan Lu, Weihua Xu, and Hongye Su

***Abstract*— To successfully implement the platoon control of connected and automated vehicles (CAVs), the model uncertainties, external disturbances, and time delays must be addressed. In this study, we propose a distributed Model Predictive Control scheme to achieve offset-free tracking for CAVs with model uncertainties and mixed disturbances which contains both stochastic and deterministic noises. A simple Vehicle-to-Vehicle communication flow with the short-distance and low-volume information exchange is adopted to reduce the effect of the communication delay and dropout. First, the deterministic and stochastic perturbances are addressed separately in constraints handling with an integrated control law based on the mismatched prediction model. Additional constraints on coupling input and output are imposed to guarantee the assumption about forecast trajectory. Then an observer gives an unbiased estimate of the state and integrated disturbance. Based on the estimation, the target calculator eliminates the steady tracking offset by designing a new target in real-time and the Min-Max Model Predictive Control steers the uncertain systems to the new target under the worst case within the domain of disturbances. Lastly, the Optimal Control Problem is reformulated and can be efficiently solved by existing solvers. Simulation results indicate that the proposed method shows superiority in interference rejection and offset-free tracking.**

***Index Terms*— Platoon control, CAV, stochastic and deterministic disturbance, model uncertainty.**

I. Introduction

RESEARCH on connected and automated vehicles(CAVs) has received much attention due to its potential to disrupt mobility, extending what is possible with driving automation and connectivity alone. A recent publication of [1] gives an insightful review of the control and planning architecture for CAVs. Among the many technologies concerned, the platoon control of the longitudinal motion is an important topic that allows vehicles to achieve system-wide benefits for energy efficiency and road throughputs.

The main objective of the longitudinal platoon control is generally accepted as regulating a chain of vehicles to an ideal steady state, where the vehicles move at the same speed while maintaining a safe and short distance between any consecutive followers. [2]

Manuscript received 26 September 2020; revised 28 June 2021; accepted 19 January 2022. Date of publication 22 March 2022; date of current version 11 October 2022. This work was supported in part by the Natural Science Foundation of Zhejiang, China, under Grant LQ21F030007, in part by the National Key Research and Development Program of China under Grant 2018YFB1701102, and in part by the National Natural Science Foundation of P.R. China, under Grant NSFC: 62073286. The Associate Editor for this article was Y. Wang. *(Corresponding author: Lei Xie.)*

Xiaorong Hu, Lei Xie, Weihua Xu, and Hongye Su are with the State Key Laboratory of Industrial Control Technology, Zhejiang University, Hangzhou 310027, China (e-mail: xiaoronghu@zju.edu.cn; leix@iipc.zju.edu.cn).

Lantao Xie is with Huawei Technologies Company Ltd., Shenzhen 518055, China.

Shan Lu is with Shenzhen Polytechnic, Shenzhen 518055, China.

Digital Object Identifier 10.1109/TITS.2022.3153307

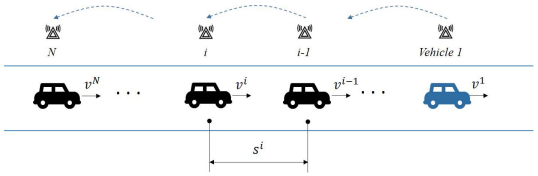


Fig. 1. Longitudinal motion of a platoon of N vehicle.

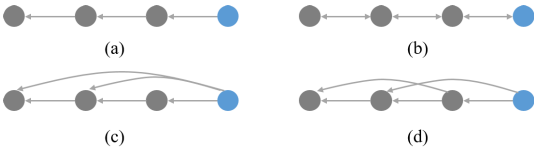


Fig. 2. Information flow network, (a) Predecessor Following, (b) Bidirectional, (c) Predecessor following leader, (d) Two predecessor following.

The distributed control schemes are suitable for the platoon problem without using centralized communication and large-scale problem computation, hence much research has been devoted to multi-agent consensus schemes. From the perspective of multi-agent consensus control, the reference [3] has done a detailed survey and decomposes the platoon control problem into four interrelated parts.

1) node dynamics, i.e. the dynamics of each CAV.

2) information flow network, i.e. the topology and quality of information flow, and the type of information exchanged. Figure 2 depicts some typical communication topologies used in platooning. The information exchanged may just be the current velocity and acceleration, or include forecasts thereof.

3) distributed controller.

4) geometry formation, i.e. vehicle ordering, cruising speed, and inter-vehicle distance.

Based on the previous work on CAVs, Model Predictive Control (MPC) has been proven to be one of the most powerful design approaches, providing a systematic framework to include forecasts and handle constraints in real-time [4]–[8]. Successful MPC applications depend on the accuracy of the process models. Uncertainties (including uncertain disturbances and model mismatches) may cause control performance to be sluggish, overly conservative, or even unstable [9].

In platooning, the uncertainty is an inevitable element given the difficulty in collecting exact information of a large number of vehicles and the varying environmental parameters. The physical factors in the real world suggest that the uncertainties could be encountered by every node and supposing that only

1558-0016 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

the leader vehicle faces uncertain traffic conditions is not appropriate. The uncertainty could not only bring a performance deterioration to the local vehicle but also propagate along the vehicle string through the forecast communication and lead to an accident. Generally, the slopes along a route result in a significant part of the disturbances [10]. The air drag variations effected by the intra-platoon spacing bring disturbances, especially for the follower vehicles [11]. Sensor inaccuracies should also be taken into consideration. As stated in [12], low-precision sensors have a negative impact on the behavior of the platoon. Besides, the mass of the vehicles and the road friction under different weather conditions are also uncertain [13].

Assuming the disturbance to be stochastic noise or deterministic noise are two typical choices in controller designing [14]. The stochastic approaches assume a known distribution of stochastic noise and allow for the easy propagation of the uncertainty through the system dynamics [15]–[17]. The defect follows as the exact distribution is rarely available in practice. In robust approaches the noises are assumed to be deterministic and lie in a bounded set, thus are more conservative without utilizing the probability characteristics [7]. In order to make the best use of the known information about the noises, we regard the external disturbances as a superposition of stochastic noise and deterministic noise. Besides, the model uncertainty could also be viewed as an equivalent deterministic noise on the nominal system in the implicit method. To sum up, we regard the effects of uncertain environment and plant as a mixture process perturbance, consisting of stochastic perturbance whose probability characteristics are retrievable and deterministic perturbance whose bound is known. The stochastic perturbance accounts for the influence of the stochastic noise on the air drag variation. The deterministic perturbance accounts for the influence of both unknown parametric model uncertainty and the original deterministic noise on road slope and mass variation. Measurement inaccuracies of on-board sensors will also be taken into account. The research in CAVs control is still in an embryonic stage with lots of challenges. When the problem involves parametric uncertainties, most existing platoon approaches turn to other methods such as the adaptive slide mode control in [18] and the barrier-function-based scheme in [19]. A recent publication [7] based on MPC uses a Min-Max formulation, but it is still a centralized control method, requiring the leader to collect/send information from/to all followers, which needs large-volume communication over a long distance.

In addition, from the overall perspective of the platoon, the delays and dropouts in communication are also factors of concern [7], [16], [20]. When more vehicles join the platoon, the communication distance gets longer and additional disturbances may be introduced in the control system especially under complex information flow types. A simpler communication scheme, Predecessor Following (PF) type is employed in this study. The field test by Ploeg *et al.* [21] suggests that the communication delay is typically very small (about 0.01 s) in a small platoon. Therefore, it's reasonable to assume that the communication process between two consecutive vehicles, is completed in one sampling interval T ($T>0.01$s). Under limited communication conditions, the difficulty of the control problem is not only to deal with the interference of local vehicle and achieve unbiased tracking to the desired velocity, but also to handle the information uncertainty from the forecast of the preceding vehicle and achieve unbiased tracking to the desired distance. Thus, the initial problem now becomes a tracking problem in a distributed MPC manner for subsystems with mixed disturbances.

Motivated by the works discussed above, this paper attempts to propose a distributed MPC scheme under a simple V2V communication flow for CAVs with mixed disturbances and model uncertainties to achieve offset-free tracking performance. The platoon are considered as a system consisting of several state-coupled subsystems with chance constraints. A combined state-feedback and disturbance-affine feedback control scheme has been incorporated to handle the constraints based on the stochastic and deterministic noises separately for less conservativeness. Similarity constraints on message are imposed since the distributed controllers work in parallel manner. To realize offset-free tracking and interference suppression, a combination of the target calculator and the min-max objective function is proposed. The target calculator is employed to eliminate the effect of disturbances tending to a steady value, which account for most local deterministic perturbances. The min-max objective function is used to address the perturbances varying over predictive horizon, which accounts for the forecast uncertainties and model uncertainties. The final Optimal Control Problem (OCP) can be reformulated by applying Schur Complement and Semidefinite Relaxation techniques to obtain tractable solutions.

The main contributions of this study are twofold: (1) We develop a distributed MPC method for platoon control and achieve offset-free objective tracking under the condition that every vehicle could be affected by mixed disturbances and model mismatch. The effect of disturbances is considered in the formulation of the predictive model to avoid violating constraints. The offset elimination considering deterministic disturbances and model mismatch are emphasized in controller designing. (2) We design a constraint formulation for the mixed disturbance in a distributed manner. The stochastic and deterministic components are treated separately when tightening the chance constraints. The stochastic disturbances are considered from a global perspective while the deterministic disturbances from a local viewpoint by lumping the error of coupling input into the local disturbances under a necessary assumption. Extra constraints on coupling output are designed to guarantee the assumption.

The paper is organized as follows. In Section 2, the problem is formally stated. Section 3 presents the constraints handling strategy and Section 4 describes the proposed distributed model predictive control algorithm. A simulation result is discussed in Section 5. Finally, in Section 6, some conclusions are drawn.

II. Problem Statement

This section discusses the platoon problem formation. The platoon is comprised of M vehicles and each vehicle is a

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

subsystem. A second-order model with mixed disturbances is built to represent the longitudinal dynamics of the vehicle. We assume that the controller vehicles are V2V enabled under the predecessor following communication type. The velocity forecast and reference are contained in the transmitted message. Considering that the individual models are dynamically coupled, the uncertainty contained in the message could propagate along the vehicle stream. The goal of the controller designing in the following sections is suppressing the propagating perturbances and eliminating the offset.

A. Assumptions and Notation

To properly define our problem, we make the following assumptions.

Assumption 1: The controlled vehicles are equipped with sensors and can measure the absolute velocity of the controlled vehicle and the relative distance to the preceding vehicle.

Assumption 2: Vehicles in the platoon can share individual information (i.e. velocity forecast) via V2V communications within a short-range communication distance. The communication between two consecutive vehicles is completed in one sampling interval T. Packet loss is not considered.

Assumption 3: Vehicles can compute and implement control decisions simultaneously.

Assumption 4: The control delay is negligible: i.e., vehicles can implement the control instantly.

Assumption 5: Vehicles in the platoon are fully automated, so that human driver characteristics are not considered.

Assumption 6: The control decision is determined at a short, constant time interval Δt.

Assumption 7: The problem focuses on the cooperating longitudinal dynamics of the controlled vehicles in a single lane. Lane-changing automation is not considered.

Notations: $\mathcal{I}_{n_1:n_2}$ is the set of integers from n_1 to n_2. Positive definite and semidefinite matrices are indicated as $X \succ 0$ and $X \succeq 0$, respectively. For an event Ev we denote the probability of occurrence as $Pr\{Ev\}$. For a random variable ω we denote the expected value as $\mathbb{E}(\omega)$ and its variance as $var(\omega)$. For a vector x, $\|x\|_Q = x^T Qx$ denotes the weighted 2-norm while $\|x\|_1$ denotes ℓ_1 norm.

Indices:

i, controlled vehicle index, i=1,2. . .M. When i=1, it refers to the leader vehicle, positioned at the front of the platoon also denoted by the index L.

k, control time index, k=1,2,3, . . .

N, prediction horizon length.

t, prediction horizon time index, t=1,2. . .N.

B. Platoon Formulation

We model the longitudinal dynamics of the ith vehicle as

$$\dot{s}^i = v^{i-1} - v^i \tag{1a}$$

$$\dot{v}^i = \frac{1}{M}\left(\frac{T_w^i}{R_w} - F_f^i\right) \tag{1b}$$

where v^i, s^i are the absolute velocity and the inter-vehicle distance, respectively, and the wheel torque T_w^i works as the manipulated input, for $i \in \mathcal{I}_{2:M}$. The index $(i-1)$ here refers to the preceding vehicle. F_f^i is the frictional force, consisting of the gravity force, the rolling resistance, and the aerodynamic drag, given by

$$F_f^i = Mg(\sin\theta + c_r\cos\theta) + \frac{1}{2}\rho A c_x (v^i)^2 \tag{2}$$

TABLE I
MODEL PARAMETERS

M	vehicle mass
R_w	wheel radius
g	gravitational constant
θ	road slope
A	vehicle reference area
ρ	air density
C_x	vehicle drag coefficient
C_r	vehicle rolling coefficient
T	sampling time

The meaning of the model parameters is shown in Table I. The velocity, distance, and wheel torque are bounded as

$$vmin \le v^i \le vmax \tag{3a}$$

$$smin \le s^i \tag{3b}$$

$$umin \le u^i \le umax \tag{3c}$$

$$\Delta umin \le \Delta u^i \le \Delta umax \tag{3d}$$

Eq.(3a)(3b) are the state constraints to regulate the maximum and minimum deviation from the equilibrium velocity and spacing, respectively. Eq.(3c)(3d) are used to guarantee the control inputs and input slew rates are within a reasonable range. For clarity, Eq.(3a)(3b) are uniformly expressed as $Gx \le g$, Eq.(3c) as $Fu \le f$ in the following constraints handling. Compared to the position-velocity model, such state-coupled model formulation (1) converts the coupling constraints on the inter-vehicle distance to local constraints.

In practice, the mass variation of vehicles, the uncertainties of environmental parameters due to the weather condition, and the incorrect slope data bring about the model mismatch between the model and the real system. To handle the effect of model mismatch and disturbances, a direct way is treating the friction term as an external disturbance and the model becomes linear naturally. However, it is hard to determine the boundary since several parameters involved are accompanied by noises. An alternative is modeling the disturbances specifically. Here the nonlinear dynamic(1) is first linearized at a work point and then discretized with the time step T. The mass variation(M) and the road slope(θ) act as deterministic noises while the aerodynamic coefficient(ρ) acts as stochastic noise. The models except the leader vehicle can be described by

$$x_{k+1}^i = A_{ii}x_k^i + B_i u_k^i + A_{i,i-1}x_k^{i-1} + B_{p,i}p_k^i + B_{q,i}q_k^i + m_e^i \tag{4}$$

where $x^i = \begin{pmatrix} s^i - \overline{s} \\ v^i - \overline{v} \end{pmatrix}$, $u^i = T_w^i - \overline{T}_w$, $p^i = \begin{pmatrix} M^i - \overline{M} \\ \theta - \overline{\theta} \end{pmatrix}$, $q^i = C_x - \overline{C}_x$, and m_e^i acts as a compensation for the linearization residual errors or other unmodeled errors.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

For representation convenience, let $d_{e,k}^i = B_{p,i}p_k^i + m_e^i$, $\omega_k^i = B_{q,i}q_k^i$. Thus we obtain

$$\Sigma_1 : x_{k+1}^1 = A_{11}x_k^1 + B_1u_k^1 + d_{e,k}^1 + \omega_k^1 \tag{5a}$$

$$\Sigma_i : x_{k+1}^i = A_{ii}x_k^i + B_iu_k^i + A_{i,i-1}x_k^{i-1} + d_{e,k}^i + \omega_k^i \tag{5b}$$

where $i \in \mathcal{I}_{2:M}$, and ω_k^i is a zero-mean white noise with variance $\overline{W}^i$. We assume ω_k^i and ω_k^j are uncorrelated for all $j \neq i$. For the leader vehicle, the inter-vehicle distance s^1 actually makes no sense, however, the matrix $A_{i,i-1}$ is required to be square to handle stochastic noise. So we retain the state s^1 and assume it equals to zero without any noise. Thus, the global dynamic for a platoon with M vehicles can be expressed with Eq.6,where the blocks $A_{11} = [0\ 0; 0\ 1]$, $A_{ii} = [1\ \ -T; 0\ 1]$, $A_{i,i-1} = [0\ T; 0\ 0]$, for $i = 2, \ldots M$; $B_i = [0\ T/\overline{M}R_w]^T$, $B_{\omega i} = B_{di} = I$, for $i = 1, \ldots M$; the upper index o refers to the global variable. We say that for subsystem i, subsystem j is its in-neighbor if and only if $A_{i,j} \neq 0$, $j \neq i$; subsystem j is its out-neighbor if and only if $A_{j,i} \neq 0$, $j \neq i$. Or, more specifically, for subsystem $\Sigma_i, i \in \mathcal{I}_{2:M-1}$, its out-neighbor is subsystem Σ_{i+1}; its in-neighbor is subsystem Σ_{i-1}. Each subsystem Σ_i is subjected to a set of local probabilistic chance constraints and hard constraints

$$P_r\{Gx_{k+t|k}^i \leq g\} \geq 1 - \varepsilon_x, \forall i \in \mathcal{I}_{1:M} \tag{7a}$$

$$P_r\{Fu_{k+t|k}^i \leq f\} \geq 1 - \varepsilon_u, \forall i \in \mathcal{I}_{1:M} \tag{7b}$$

$$F_hu_{k+t|k}^i \leq f_h, \forall i \in \mathcal{I}_{1:M} \tag{7c}$$

where ε_x and ε_u are the levels of constraint satisfaction for states and inputs. In the real world, the stochastic noise on aerodynamic coefficient is bounded and can be regarded as the normal distribution truncated to $\pm 2\sigma$ or $\pm 3\sigma$, which correspond to different choices of the satisfaction level.

C. Vehicle-to-Vehicle Communication

Cooperative driving in the platoon is enabled by equipping each node with vehicle-to-vehicle communication capabilities. The main content of information depends on the requirements of vehicle dynamic and further prediction. Here the forecasted velocity of the preceding vehicle is transmitted. Messages sent from vehicle i at current control time k takes the following form

$$m_k^i = \left[t_{sent};\ v_{k|k}^i;\ v_{k+1|k}^i;\ \cdots\ v_{k+N|k}^i;\ v_{des}^i\right] \tag{8}$$

where t_{sent} is the universal time and the time step for communication is the same as the sampling time used for MPC. v_{des}^i should be transmitted with the forecast because in PF flow the following vehicle cannot directly obtain the desired reference of leader vehicle.

Notice that the information exchanged also contains uncertainty. The disturbances may be amplified when propagated downstream along the vehicle string and the inter-vehicle distance fluctuates. A common approach is transmitting the leader's forecast to every follower vehicle to improve the overall performance, i.e. the predecessor-following leader (PFL) flow. All the followers set targets according to the leader's forecast, hence the maneuvers of the leader vehicle are valued. In this case, the platoon needs large-volume communication over a long distance, often resulting in delays and dropouts. On the contrary, the predecessor following flow greatly reduces the burden of communication and has advantages in dealing with dynamics heterogeneity [22]. However, once any of the vehicles in the queue have a tracking error, the other vehicles cannot adjust accordingly due to limited delivery information. Therefore, the PF type puts forward higher requirements for the controllers in offset-free tracking.

We construct the predecessor following flow in the platoon and improve the global control performance by guaranteeing the offset-free tracking between any two consecutive vehicles. The distributed manner is built as the parallel scheme where the local controller solves the problem simultaneously. This approach essentially assumes that the forecast of the preceding vehicle (last adopted) will remain the same in the future. To support this assumption, a constraint is imposed to restrict the absolute value of the difference between the message sent at the current time t_{sent} and the last moment $(t_{sent} - \Delta t)$, that is

$$|v_{k+t|k}^i - v_{k+t|k-1}^i| \leq t \cdot \tau_m, \forall t \in \mathcal{I}_{1:N-1} \tag{9}$$

where τ_m is a constant designed by the user. The idea behind increasing tolerance with the forecast horizon is that each vehicle could change the future desire point but couldn't alter the current manipulation rapidly. For the sake of uniform expression, the message constraints are rewritten as $G_hx_{k+t|k} \leq g_h$.

D. Platoon Objective

The global objective of the platoon is to track the speed of the leader while maintaining the desired distance between any pair of adjacent vehicles, specifically described as

$$\lim_{t\to\infty} \|v_t^i - v_{des}\| = 0, \quad \forall i \in \mathcal{I}_{1:M} \tag{10a}$$

$$\lim_{t\to\infty} \|s_t^i - s_{des}\| = 0, \quad \forall i \in \mathcal{I}_{2:M} \tag{10b}$$

The inter-vehicle spacing policy, which means the choice of a reference relative distance s_{des} between two consecutive vehicles determines the formation geometry of a platoon. Two major policies are adopted in most literature: 1) constant distance policy, s_{des} is a given constant positive number; 2) constant time headway policy, $s_{des}^i = s_c + t_hv^i$ where t_h is

$$x_{k+1}^o = \begin{pmatrix} A_{11} & 0 & \cdots & 0 \\ A_{21} & A_{22} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & A_{M,M-1} & A_{MM} \end{pmatrix} x_k^o + \begin{pmatrix} B_1 & 0 & \cdots & 0 \\ 0 & B_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & B_M \end{pmatrix} u_k^o + \begin{pmatrix} B_{\omega 1} & 0 & \cdots & 0 \\ 0 & B_{\omega 2} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & B_{\omega M} \end{pmatrix} \omega_k^o + \begin{pmatrix} B_{d1} & 0 & \cdots & 0 \\ 0 & B_{d2} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & B_{dM} \end{pmatrix} d_{e,k}^o \tag{6}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

the time headway, and s_c is a positive number. The constant time headway policy is in accord with driver behaviors to some extent and helps to improve the string stability, thus it is adopted in this paper.[3]

From (10) and the specified formation geometry, we could derive the reference point of vehicle i as

$$r_k^i = (s_{des}^i, v_{des}^i, u_{des}^i) \tag{11}$$

where $v_{des}^i = v_{des}^{i-1}$ is obtained from the message, $s_{des}^i = s_c + t_h v_k^i$ is determined by the current speed, $u_{des}^i = u_{ss}$ counterbalances the friction term in steady state. Thus, the cost function J_k^i is defined as

$$J_k^i = q_1 \cdot \Sigma_{t=1}^N (\Delta s_t)^2 + q_2 \cdot \Sigma_{t=1}^N (\Delta v_t)^2 + r \cdot \Sigma_{t=0}^{N-1} (\Delta u_t)^2 \tag{12}$$

where $\Delta s_t = s_{k+t|k}^i - s_{des}^i$, $\Delta v_t = v_{k+t|k}^i - v_{des}^i$, and $\Delta u_t = u_{k+t|k}^i - u_{des}^i$. The first term implies that the vehicles tend to reach the desired gap. The second term works by making the following vehicle track the same desired speed. The control cost penalizes engine operation to benefit in fuel consumption and ride comfort. As for the leader vehicle, the first term is neglected.

The control objective for each CAV is described as minimizing the cost function J_k^i while the system states and control variables are limited by constraints (3) and (9). On account of the mixed disturbance, the control problem cannot be directly solved. The following sections present constraints handling strategy and model reformulations to obtain tractable solutions that can be applied in real-time. Besides, the steady-state error in consideration of the model mismatch and external disturbances influences the distance tracking performance. The benefits of the platoon in road transportation will be discounted due to the tracking error. Therefore, offset-free tracking will be emphasized in the design of the local controller.

III. Constraint Handling Strategy

The chance constraints in (7) cannot be directly used in a optimization problem. In this section, the constraints will be reformulated into a deterministic form. The mismatch model will be used to predict the error propagating and further to tighten the constraints. For clarity, the global dynamic of the platoon (6), shown at the bottom of the previous page is rewritten as

$$x_{k+1}^o = A^o x_k^o + B^o u_k^o + \omega_k^o + d_{e,k}^o \tag{13}$$

The additive stochastic noise ω_k^o is assumed to follow a multivariate normal distribution $\mathcal{N}(0, \overline{W})$ and has unbounded support. The global covariance matrix $\overline{W} = diag(\overline{W}^1, \overline{W}^2, \ldots \overline{W}^M)$. The deterministic noise $d_{e,k}^o$ is assumed to lie in the bounded domain $\mathcal{D} = \mathcal{D}^1 \times \mathcal{D}^2 \ldots \times \mathcal{D}^M$. As mentioned before, in the following sections about controller designing we use deterministic perturbance to indicate the influence of both the original deterministic noises and the model errors to distinguish it from the external deterministic noise. For consistency, stochastic perturbance represents the influence of the stochastic noise. The basic idea of the controller scheme is dealing with deterministic and stochastic perturbances separately.

A. Handling Stochastic Perturbance ω_k^i

We start with the stochastic component by splitting the global state into $x_k^o = z_k^o + e_k^o$. e_k^o denotes the error state caused by stochastic noise, while z_k^o denotes the system state without stochastic noise. When the input takes the state-feedback form $u^o = Kx^o$, the dynamic of the global error is $e_{k+1|k}^o = (A^o + B^o K)e_{k|k}^o + \omega_k^o$, with $\mathbb{E}(e_{k+t|k}^o) = 0$, $var(e_{k+t|k}^o) = \mathbb{E}[e_{k+t|k}^o (e_{k+t|k}^o)^T]$, $\forall t \in \mathcal{I}_{1:N}$. Here we use the shorthand notation for $var(e_{k+t|k}^o) = W_{k+t|k}$. The covariance iterates as $W_{k+1|k} = (A^o + B^o K) W_{k|k} (A^o + B^o K)^T + \overline{W}$, where $(A^o + B^o K)$ is Schur stable.

An intuitive problem is that $W_{k+t|k}$ will become full through the nonzero coupling block $A_{i,i-1}$. The off-diagonal blocks make it quite complex for the local controller to detach the stochastic error. This issue can be addressed using the concept of block-diagonal upper bounding matrix as references [23] and [24]. Define a symmetric and positive semi-definite block-diagonal matrix

$$\tilde{W}_k = diag(\tilde{W}_k^1, \tilde{W}_k^2, \ldots \tilde{W}_k^M) \tag{14}$$

The evolution $\tilde{W}_{k+1|k} = diag(\tilde{W}_{k+1|k}^1, \tilde{W}_{k+1|k}^2, \ldots \tilde{W}_{k+1|k}^M$ is governed by the update of local covariance matrices as the following set of equations

$$\begin{aligned}\tilde{W}_{k+1|k}^i = &(A_{ii} + B_i K_{ii}) \tilde{W}_k^i (A_{ii} + B_i K_{ii})^T + \overline{W} \\ &+ (A_{i,i-1} + B_i K_{i,i-1}) \tilde{W}_k^{i-1} (A_{i,i-1} + B_i K_{i,i-1})^T\end{aligned} \tag{15}$$

Note that the update equations are still coupled. The cascade structure helps to complete the iteration of the global upper bounding matrix easily when platooning setup.

Lemma 1: Assume that at time k, $\tilde{W}_k \succeq W_k$. Then, if $\tilde{W}_{k+1|k}$ is defined according to (15) we have $\tilde{W}_{k+1|k} \succeq W_{k+1|k}$ Moreover, since $\tilde{W}_k \succeq W_k$, then also $\tilde{W}_k^i \succeq W_k^i$.

The detailed proof of Lemma 1 refers to [25]. The covariance of the input can be defined as $U_k^i = K_i W_k^i K_i^T, \forall i \in \mathcal{I}_{1:M}$. Correspondingly, the bounding matrices are defined as

$$\tilde{U}_k^i = K_i \tilde{W}_k^i K_i^T \tag{16}$$

which have the property that $\tilde{U}_k^i \succeq \tilde{U}_k^i, \forall i \in \mathcal{I}_{1:M}$.

Now consider the control law of the form $u_{k+t|k}^i = K_{ii}(x_{k+t|k}^i - z_{k+t|k}^i) + \pi_{k+t|k}^i$, where $\pi_{k+t|k}^i$ becomes the new optimization variable and the state-feedback component $K_{ii}(x_{k+t|k}^i - z_{k+t|k}^i)$ is employed to keep the prediction error in a neighborhood. Thus the prediction are given as

$$z_{t+1}^i = A_{ii} z_t^i + B_i \pi_t^i + A_{i,i-1}(z_t^{i-1} + e_t^{i-1}) + d_{e,t}^i \tag{17a}$$

$$e_{t+1}^i = (A_{ii} + B_i K_{ii}) e_t^i + \omega_t^i \tag{17b}$$

where the coupling term $A_{i,i-1} e_t^{i-1}$ equals to zero, since the off-diagonal blocks of the covariance matrix are zero. The index $(t+1)$ refers to $(k+t+1|k)$. Therefore, (17) simplify to

$$z_{t+1}^i = A_{ii} z_t^i + B_i \pi_t^i + A_{i,i-1} z_t^{i-1} + d_{e,t} \tag{18a}$$

$$e_{t+1}^i = (A_{ii} + B_i K_{ii}) e_t^i + \omega_t^i \tag{18b}$$

where $e_{k|k}^i \sim \mathcal{N}(0, \tilde{W}_k)$. As $x_{k+t|k}^i = z_{k+t|k}^i + e_{k+t|k}^i$, the chance constraints (7a)-(7b) can be reformulated as hard

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

constraints:

$$Gz^i_{k+t|k} \leq g - \beta^i_t, t \in \mathcal{I}_{1:N} \tag{19a}$$

$$F\pi^i_{k+t|k} \leq f - \alpha^i_t, t \in \mathcal{I}_{1:N} \tag{19b}$$

where the pth element of β^i_t is the solution to

$$\min_b b \quad s.t. \quad Pr\{g_p e^i_{k+t|k} \leq b\} \geq 1 - \varepsilon_x \tag{20}$$

Also, the pth element of α^i_t is the solution to

$$\min_a a \quad s.t. \quad Pr\{f_p K_{ii} e^i_{k+t|k} \leq a\} \geq 1 - \varepsilon_u \tag{21}$$

Here g_p and f_p are the pth row of G and F, respectively. and $\overline{g}_p$ is the pth element of g. The proof of the reformulation refers to [25].

The two standardized Stochastic Programming problems (20)(21) can be approximated as deterministic ones through the scenario-based method. Compared with the deterministic equivalent-based methods, it brings less conservatism without the linearization process. A reasonable sampling number is $\overline{N} = (1-\varepsilon_x)(e/(e-1))(\ln(1-\eta))$. η is the confidence parameter.

B. Handling Deterministic Perturbance d^i_k

Now we set about dealing with the deterministic noise $d_{e,k}$ under the dynamic (18a). A similar concern is that the perturbance may come from local model mismatch and message uncertainties. As a preparation, we rewrite the dynamic (18a) with coupling input and output as

$$z^i_{k+t+1|k} = A_{ii} z^i_{k+t|k} + B_i \pi^i_{k+t|k} + H_i \varpi^i_{k+t|k} + d^i_{e,k+t|k} \tag{22a}$$

$$y^i_{z,k+t|k} = C_{zi} z^i_{k+t|k} \tag{22b}$$

From (18a), given the elements in the first column of $A_{i,i-1}$ are zero, the term $A_{i,i-1} z^{i-1}$ can be replaced by $H_i \varpi^i$ where $H_i = [T;\ 0]$ is induced from $A_{i,i-1}$ by removing the first column and becomes consistent with the forecast message on the dimension. ϖ^i represent the coupling input, more specifically, the velocity variation of the preceding vehicle, i.e $\varpi^i_{k+t} = y^{i-1}_{z,k+t}$. $\tilde{\boldsymbol{\varpi}}^i_k$ represent the forecast velocity trajectory obtained from the latest preceding message m^{i-1}_{k-1}. $y^i_{z,k+t|k}$ represents the coupling output, that is, the velocity variation of the controlled vehicle. $\tilde{\boldsymbol{y}}^i_{z,k-1}$ represents the forecast velocity trajectory transmitted to the following vehicle by the message m^i_{k-1}.

As we stated before, each subsystem Σ_i transmits to the out-neighbor (subsystem Σ_{i+1}) its planned trajectory $\tilde{y}^i_{z,k}$ over the predictive horizon. An assumption about coupling input and output is necessary for the following deterministic perturbance handling, that is, the actual trajectory of each subsystem lies in a neighborhood centered in its last forecast, i.e $y^i_{z,k+t} \in \tilde{y}^i_{z,k+t|k-1} \oplus \mathcal{E}^i_t, \forall t \in \mathcal{I}_{0:N-1}, \mathcal{E}^i_t = \{\epsilon | G^i_{e,t}\epsilon \leq g^i_{e,t}\}$. This assumption can be guaranteed by directly applying constraints

$$G^i_{e,t}(C_{zi} z^i_{k+t|k} - \tilde{y}^i_{z,k+t|k-1}) \leq g^i_{e,t} \tag{23}$$

where $\tilde{y}^i_{z,k+t|k-1}$ is a given constant at the time step k. Under the assumption above, $\varpi^i_{k+t} = \tilde{\varpi}^i_{k+t} + \epsilon_{n,k+t}$, and $\epsilon_{n,k+t} \in \mathcal{E}^{i-1}_t$. Therefore the dynamic is equivalent to

$$z^i_{k+t+1|k} = A_{ii} z^i_{k+t|k} + B_i \pi^i_{k+t|k} + H_i(\tilde{\varpi}^i_{k+t|k} + \epsilon_{n,k+t|k}) + d^i_{e,k+t|k} \tag{24}$$

Different from the treatment of stochastic noise, here the message uncertainties are lumped into the local deterministic perturbance d^i_{k+t}. In this way, (22) is rewritten as

$$z^i_{k+t+1|k} = A_{ii} z^i_{k+t|k} + B_i \pi^i_{k+t|k} + H_i \tilde{\varpi}^i_{k+t|k} + d^i_{k+t|k} \tag{25}$$

Here the term $H_i \tilde{\varpi}^i_{k+t|k}$ is equivalent to a non-manipulated input, known in advance over the prediction horizon. The deterministic perturbance $d^i_{k+t|k} = d^i_{e,k+t|k} + H_i \epsilon_{n,k+t|k}$. The domains of deterministic perturbance over the predictive horizon, denoted as $d^i_{k+t|k} \in \mathcal{D}^i_{k+t}, t \in \mathcal{I}_{0:N-1}$ need to be recalculated as $D^i_{k+t} = D^i_{e,k+t} \oplus H_i D^{i-1}_{e,k+t}$. For the sake of computation, the set $\{\mathcal{D}^i\}$ will be standardized as [25]. Introduce a unit perturbance set $\mathcal{D}_u = \{d \in \mathbb{R}^{n_x} | \|d\|_\infty \leq \mathbf{1}\}$. For each $\mathcal{D}^i_{k+t}$, find the smallest diagonal weight matrix P^i_t satisfying $\mathcal{D}^i \subseteq P^i_t \mathcal{D}_u$.

A predictive control law with affine dependence on perturbances are considered here,i.e. $\pi^i_{k+t|k} = \prod_{m=0}^{t-1} L_{t,m} d^i_{k+t|k} + \nu^i_{k+t|k}$. It is a bijective transformation of the class of time-varying linear feedback plus feedforward control laws. The transformation leads to a convex optimization and renders computation tractable. After standardization, the control law can be written equivalently as $\pi^i_{k+t|k} = \prod_{m=0}^{t-1} L_{t,m} P^i_m \bar{d}^i_{k+t|k} + \nu^i_{k+t|k}$, where $\bar{d}^i_{k+t|k} \in \mathcal{D}_u$, $L_{t,m}$ and $\nu_{k+t|k}$ are decision variables. Expressed as compact notation, $\boldsymbol{\pi}^i_k = \boldsymbol{L}^i_k \boldsymbol{P}^i \bar{\boldsymbol{d}}^i_k + \boldsymbol{\nu}$, with a lower-triangular structure

$$\boldsymbol{L}^i_k = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ L^i_{1,0|k} & 0 & \cdots & 0 & 0 \\ L^i_{2,0|k} & L^i_{2,1|k} & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ L^i_{N-2,0|k} & L^i_{N-2,1|k} & \cdots & L^i_{N-2,N-1|k} & 0 \end{pmatrix}$$

and $\boldsymbol{\pi}^i_k = (\pi^i_{k|k}, \pi^i_{k+1|k}, \ldots \pi^i_{k+N-1|k})^T$, $\boldsymbol{P}^i = diag(P^i_0, P^i_1, \ldots P^i_{N-1})$, $\bar{\boldsymbol{d}}^i_k = (\bar{d}^i_{k|k}, \bar{d}^i_{k+1|k}, \ldots \bar{d}^i_{k+N-1|k})^T$, $\boldsymbol{\nu} = (\nu^i_{k|k}, \nu^i_{k+1|k}, \ldots \nu^i_{k+N-1|k})^T$. The corresponding vector of predicted state assumes the form

$$\boldsymbol{z}^i_k = \mathcal{A}_{ii} z^i_{k|k} + \mathcal{B}_i \boldsymbol{\nu}^i_k + \mathcal{H}_i \tilde{\boldsymbol{\varpi}}^i + (\mathcal{M}_i + \mathcal{B}_i \boldsymbol{L}^i_k) \boldsymbol{P}^i \bar{\boldsymbol{d}}^i_k \tag{26}$$

where $\mathcal{A}_{ii}$, $\mathcal{B}_i$, $\mathcal{H}_i$ and $\mathcal{M}_i$, respectively, denote the convolution matrices from z^i, $\boldsymbol{\pi}^i$, $\tilde{\boldsymbol{\varpi}}^i$ and $\bar{\boldsymbol{d}}^i$ to $\boldsymbol{z}^i$ given by

$$\mathcal{A}_{ii} = \begin{pmatrix} A_{ii} & A^2_{ii} & \cdots & A^N_{ii} \end{pmatrix}^T, \ \mathcal{B}_i = Mat(A_{ii}, B_i, N),$$
$$\mathcal{H}_i = Mat(A_{ii}, H_i, N), \quad \mathcal{M}_i = Mat(A_{ii}, I, N)$$

The elements $M_{i,j}$ in matrix $Mat(A, B, N)$ satisfies that $M_i j = A^{i-j} B$ if $i >= j$, otherwise the element is zero.

Substituting (26) into (19) yields

$$\mathcal{G}(\mathcal{A}_{ii} z^i_{k|k} + \mathcal{B}_i \boldsymbol{\nu}^i_k + \mathcal{H}_i \tilde{\boldsymbol{\varpi}}^i) + \max_{\bar{\boldsymbol{d}}^i_k \in \mathcal{D}_u \times \cdots \times \mathcal{D}_u} \mathcal{G}(\mathcal{M}_i + \mathcal{B}_i \boldsymbol{L}^i_k) \boldsymbol{P}^i \bar{\boldsymbol{d}}^i_k \leq \boldsymbol{g} - \beta \tag{27}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

where $\mathcal{G} = diag(G, G \ldots . G)$, $\boldsymbol{g} = (g, g, \ldots g)$, $\beta = (\beta_1^i, \beta_2^i, \ldots \beta_N^i)$. The maximum of linear function in the unit cube theorem can be used to express the constraint (27) as the equivalent linear constraints.

Theorem 1 (Maximum of Linear Function in the Unit Cube):

$$\max_{|x| \le 1} c^T x = \|c\|_1 = |c^T| \mathbf{1} \tag{28}$$

The final formulation of constraints (7a) is given by

$$\mathcal{G}(\mathcal{A}_{ii} x_{k|k}^i + \mathcal{B}_i \boldsymbol{v}_k^i + \mathcal{H}_i \tilde{\boldsymbol{\varpi}}^i) \le \boldsymbol{g} - \beta - \Psi_{k,x}^i \cdot \mathbf{1} \tag{29a}$$

$$\mathcal{G}(\mathcal{M}_i + \mathcal{B}_i \boldsymbol{L}_k^i) \boldsymbol{P}^i \le \Psi_{k,x}^i \tag{29b}$$

$$-\mathcal{G}(\mathcal{M}_i + \mathcal{B}_i \boldsymbol{L}_k^i) \boldsymbol{P}^i \le \Psi_{k,x}^i \tag{29c}$$

Proof: What have to be done is to maximize each row in the uncertain term $\mathcal{G}(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P}\boldsymbol{d}$. For clarify, the index i, k are neglected. Partition the rows of the matrix $\mathcal{G}(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P}$ as

$$\mathcal{G}(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P} = \left(\zeta_1^T, \zeta_2^T, \cdots\right)^T \tag{30}$$

From (30) and Theorem 1, the pth row of the uncertain term $\max(\mathcal{G}(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P}\boldsymbol{d})_p = \max \zeta_p^T \boldsymbol{d} = \|\zeta_p\|_1 = |\zeta_p^T| \mathbf{1}$. It follows that (27) is equivalent to

$$\mathcal{G}(\mathcal{A} z_k + \mathcal{B}\boldsymbol{v} + \mathcal{H}\tilde{\boldsymbol{\varpi}}) + |\mathcal{G}(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P}| \cdot \mathbf{1} \le \boldsymbol{g} - \beta \tag{31}$$

The constraint (31) still cannot be used directly for linear programming on account of the element-wise absolute value operation on matrix $(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P}$. The matrix variable Ψ is defined to bound the absolute value of the matrix, i.e.

$$\pm \mathcal{G}(\mathcal{M} + \mathcal{B}\boldsymbol{L})\boldsymbol{P} \le \Psi \tag{32}$$

As a result, substituting $x_{k|k} = z_{k|k}$, the constraint (31) is equivalent to

$$\mathcal{G}(\mathcal{A} x_k + \mathcal{B}\boldsymbol{v} + \mathcal{H}\tilde{\boldsymbol{\varpi}}) + \Psi \cdot \mathbf{1} \le \boldsymbol{g} - \beta \tag{33}$$

with (32). ■

Analogously, constraints (7b) is given by

$$\mathcal{F}\boldsymbol{v}_k^i \le \boldsymbol{f} - \alpha - \Psi_{k,u}^i \cdot \mathbf{1} \tag{34a}$$

$$\mathcal{F}\boldsymbol{L}_k^i \boldsymbol{P}^i \le \Psi_{k,u}^i \tag{34b}$$

$$-\mathcal{F}\boldsymbol{L}_k^i \boldsymbol{P}^i \le \Psi_{k,u}^i \tag{34c}$$

where $\mathcal{F} = diag(F, F, \cdots F,)$, $\boldsymbol{f} = (f, f, \cdots f)$, $\alpha = (\alpha_1^i, \alpha_2^i, \ldots \alpha_N^i)$.

As for constraints (23), it is given by

$$\mathcal{G}_e^i \mathcal{C}_{zi} (\overline{\mathcal{A}}_{ii} x_{k|k}^i + \overline{\mathcal{B}}_i \boldsymbol{v}_k^i + \overline{\mathcal{H}}_i \tilde{\boldsymbol{\varpi}}^i) \le \boldsymbol{g}_e^i + \mathcal{G}_e^i \tilde{\boldsymbol{y}}_{z,k-1}^i - \Psi_{k,e}^i \cdot \mathbf{1} \tag{35a}$$

$$\mathcal{G}_e^i \mathcal{C}_{zi} (\overline{\mathcal{M}}_i + \overline{\mathcal{B}}_i \boldsymbol{L}_k^i) \boldsymbol{P}^i \le \Psi_{k,e}^i \tag{35b}$$

$$-\mathcal{G}_e^i \mathcal{C}_{zi} (\overline{\mathcal{M}}_i + \overline{\mathcal{B}}_i \boldsymbol{L}_k^i) \boldsymbol{P}^i \le \Psi_{k,e}^i \tag{35c}$$

where $\mathcal{G}_e^i = diag(G_{e,0}^i, G_{e,1}^i, \cdots G_{e,N-1}^i)$, $\mathcal{C}_{zi} = diag(C_{zi}, C_{zi}, \cdots C_{zi})$, $\boldsymbol{g}_e^i = (g_{e,0}^i, g_{e,1}^i, \cdots g_{e,N-1}^i)$, $\Psi_{k,u}^i$, $\Psi_{k,x}^i$, $\Psi_{k,e}^i$ $\boldsymbol{L}_k^i$, $\boldsymbol{v}_k^i$ are decision variables at each time step k, and the matrices are given by

$$\overline{\mathcal{A}}_{ii} = \left(I \;\; A_{ii} \;\; \cdots \;\; A_{ii}^{N-1}\right)^T, \quad \overline{\mathcal{B}}_i = \overline{Mat}(A_{ii}, B_i, N),$$
$$\overline{\mathcal{H}}_i = \overline{Mat}(A_{ii}, H_i, N), \quad \overline{\mathcal{M}}_i = \overline{Mat}(A_{ii}, I, N)$$

The elements $M_{i,j}$ in matrix $\overline{Mat}(A, B, N)$ satisfies that $M_i j = A^{i-j-1} B$ if $i >= j + 1$, otherwise the element is zero.

IV. The Distributed Model Predictive Control Scheme to Mixed Disturbances

In this section, we develop an optimization problem to determine the control command. Under the preceding-follower flow, offset-free control is the key point to guarantee global performance in platooning. If the uncertainty signal d_k^i tends to a steady value d_∞^i, for example, a long-distance uphill scenario, then, the system may exhibit a steady tracking offset $y_\infty^i - r_k^i$, where y_∞^i is the steady output, and r_k^i is the reference point. One can minimize it by designing a new artificial target (x_s, u_s) taking the steady-state error into consideration. To remove the offset on-line, an estimation of perturbance must be calculated by means of an appropriate observer. The Min-Max Model Predictive Control could steer the uncertain subsystems to the new target under the worst case. Note that the offset-free tracking here refers to the expectation of the outputs rather than an absolute zero due to the stochastic noise. Now that the state decomposition and model reformulation in Section 3 have transformed the individual predictive models into dynamically decoupled ones by splitting the global stochastic error and lumping the forecast uncertainties into local disturbance, we omit the index i in some notation here for clarity.

A. Target Calculator Formulation

An augmented model is introduced to achieve an unbiased and efficient estimation of the state and integrating perturbance. The perturbance d_k here is a new state and the model is described as follow:

$$\begin{pmatrix} x \\ d \end{pmatrix}_{k+1} = \begin{pmatrix} A & B_d \\ 0 & I \end{pmatrix} \begin{pmatrix} x \\ d \end{pmatrix}_k + \begin{pmatrix} B & H \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ \tilde{\varpi} \end{pmatrix}_k + \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} \omega \\ \xi \end{pmatrix}_k \tag{36a}$$

$$y_k = \begin{pmatrix} C & C_d \end{pmatrix} \begin{pmatrix} x \\ d \end{pmatrix}_k + \eta_k \tag{36b}$$

where $\xi_k \in \mathbb{R}^{n_d}$ is a zero-mean white noise and $\eta_k \in \mathbb{R}^{n_y}$ is the output noise. A steady-state Kalman filter is designed based on the augment model. Once the new measurement data y_k^m is obtained, the correction is given by

$$\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix}_{k|k} = \begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix}_{k|k-1} + \begin{pmatrix} L_x \\ L_d \end{pmatrix} (y_k^m - C\hat{x}_{k|k-1} - C_d \hat{d}_{k|k-1})$$

Besides, the prediction is given by

$$\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix}_{k+1|k} = \begin{pmatrix} A & B_d \\ 0 & I \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix}_{k|k} + \begin{pmatrix} B & H \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ \tilde{\varpi} \end{pmatrix}_{k|k}$$

$$\hat{y}_{k+k} = \begin{pmatrix} C & C_d \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix}_{k|k}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

Here we use a hat to distinguish the observed value. The gain matrices for states and perturbances, L_x and L_d, are solved by a relative discrete algebraic Riccati equation. The matrix B_d and C_d would be designed by the user, and the following equations should be satisfied when selecting to make (36) detectable and remove offset:

$$\operatorname{rank}\begin{pmatrix} I-A & -B_d \\ C & C_d \end{pmatrix} = n_x + n_d;\ \operatorname{rank}(L_d) = n_d = n_y;$$

With the knowledge of the estimated perturbance $\hat{d}_k$ and a given reference point r_k, any steady state of the system (x_s, u_s) must satisfy the following equations.

$$x_s = Ax_s + Bu_s + H\varpi_s + B_d\hat{d}_k \tag{37a}$$
$$r_k = Cx_s + Du_s + C_d\hat{d}_k \tag{37b}$$

It is hard to find the solution for every $\hat{d}_k$ and ϖ_s, where ϖ_s is the desired state of the preceding vehicle. An alternative is to perform an additional optimization problem like

$$\min \quad e_s^T Q_s e_s + u_s^T R_s u_s \tag{38a}$$
$$s.t. \quad x_s = Ax_s + Bu_s + H\varpi_s + B_d\hat{d}_k \tag{38b}$$
$$e_s = Cx_s + Du_s + C_d\hat{d}_k - r_k \tag{38c}$$
$$Gx_s \le g \tag{38d}$$
$$Fu_s \le f \tag{38e}$$

where e_s admits an error between the observed output y_k and the reference r_k.

B. Min-Max Cost Function

The target calculator focuses on the case where d_k^i tends to a steady value d_∞^i. Thus the modified target together with a quadratic cost function of the nominal system as (12) could efficiently achieve offset cancellation for an independent system. However, in the platoon problem, the vehicles have coupling dynamics. Not only the disturbances acting on the subsystem itself but also the uncertainties contained in the imprecise forecast from the preceding one influence the control performance. Thus the varying perturbance should be further addressed. Natural thinking is adopting the cost function as a Min-Max form. The concept of minimizing the worst case is applied, that is, an optimal solution is generated to minimize the largest cost value of J when the perturbance $d_{k+t|k}^i$ is used to predict the future states. Besides, the cost function should be integrated with the expectation form in the stochastic framework. Consequently, the initial cost function (12) for each subsystem is reformulated as

$$\min_{\pi_{k+t}^i}\max_{\bar{d}_{k+t}^i}\sum_{t=0}^{N-1}\|\mathbb{E}(r_{k+t}^{i,c}) - r_{k+t}^{i,c}\|_Q + \|\mathbb{E}(u_{k+t}^i) - u_s^i\|_R + \|\mathbb{E}(r_{k+N}^{i,c}) - r_{k+N}^{i,c}\|_P \tag{39}$$

where $Q = diag(q_1, q_2)$, $R = r$. Notation $r^{i,c}$ represents the controlled output of the real system distinguished with the model output y^i. According to the unbiased observer before, the expectation satisfies $\mathbb{E}(r_{k+t}^i) = \mathbb{E}(\hat{y}_{k+t}^i) = C^i \cdot \mathbb{E}(\hat{x}_{k+t}^i) = \mathbb{E}(\hat{x}_{k+t}^i) \doteq \bar{x}_{k+t}^i$, $\mathbb{E}(u_{k+t}^i) = \pi_{k+t}$. Then (39) is equivalent to

$$\min_{\pi_{k+t}^i}\max_{\bar{d}_{k+t}^i}\sum_{t=0}^{N-1}\|\bar{x}_{k+t}^i - x_s^i\|_Q + \|\pi_{k+t}^i - u_s^i\|_R + \|\bar{x}_{k+N}^i - x_s^i\|_P \tag{40}$$

where $\bar{x}_{k+1|k}^i = A_{ii}\bar{x}_{k|k}^i + B_i\pi_{k|k}^i + \hat{d}_{k|k}^i + H^i\tilde{\varpi}_{k|k-1}^i$. Such a problem cannot be solved directly.

The vectors and matrices mentioned in Section 3 enables the compact notation of (40). Introduce the epigraph form of the optimization problem and define the problem as

$$\min_{\boldsymbol{\pi}} c$$
$$s.t.\ (\bar{\boldsymbol{x}} - \boldsymbol{x_s})^T \boldsymbol{Q}(\bar{\boldsymbol{x}} - \boldsymbol{x_s}) + (\boldsymbol{\pi} - \boldsymbol{u_s})^T \boldsymbol{R}(\boldsymbol{\pi} - \boldsymbol{u_s}) \le c, \quad \forall \bar{\boldsymbol{d}}$$

Applying as Schur complement on the constraint takes us to the matrix space.

$$\begin{pmatrix} c & (\bar{\boldsymbol{x}}-\boldsymbol{x_s})^T & (\boldsymbol{\pi}-\boldsymbol{u_s})^T \\ (\bar{\boldsymbol{x}}-\boldsymbol{x_s}) & \boldsymbol{Q}^{-1} & 0 \\ (\boldsymbol{\pi}-\boldsymbol{u_s})^T & 0 & \boldsymbol{R}^{-1} \end{pmatrix} \succeq 0 \tag{42}$$

Substitute $\bar{\boldsymbol{x}} = \mathcal{A}\bar{x}_{k|k} + \mathcal{B}\boldsymbol{v} + \mathcal{H}\tilde{\boldsymbol{\varpi}} + \mathcal{B}\boldsymbol{LP}\bar{\boldsymbol{d}} + \boldsymbol{B_d}\hat{\boldsymbol{d}}$, $\boldsymbol{\pi} = \boldsymbol{LP}\bar{\boldsymbol{d}} + \boldsymbol{v}$. Separate certain and uncertain terms, and apply Semidefinite Relaxations. We obtain a sufficient condition for (42) to hold.

$$\begin{pmatrix} c-\|\tau\|_1 & (\bar{X}-\boldsymbol{x_s})^T & (\boldsymbol{\pi}-\boldsymbol{u_s})^T & 0 \\ (\bar{X}-\boldsymbol{x_s}) & \boldsymbol{Q}^{-1} & 0 & \mathcal{B}\boldsymbol{LP} \\ (\boldsymbol{\pi}-\boldsymbol{u_s})^T & 0 & \boldsymbol{R}^{-1} & \boldsymbol{LP} \\ 0 & (\mathcal{B}\boldsymbol{LP})^T & (\boldsymbol{LP})^T & diag(\tau) \end{pmatrix} \succeq 0 \tag{43}$$

where $\bar{X} = \mathcal{A}\bar{x}_{k|k} + \mathcal{B}\boldsymbol{v} + \mathcal{H}\tilde{\boldsymbol{\varpi}} + \boldsymbol{B_d}\hat{\boldsymbol{d}}$.

The Optimal Control Problem(OCP) can be expressed as:

$$\Omega = \min_{\boldsymbol{L},\boldsymbol{v},c,\tau,\Psi} c \tag{44a}$$
$$s.t. \quad \begin{pmatrix} c-\|\tau\|_1 & (\bar{X}-\boldsymbol{x_s})^T & (\boldsymbol{\pi}-\boldsymbol{u_s})^T & 0 \\ (\bar{X}-\boldsymbol{x_s}) & \boldsymbol{Q}^{-1} & 0 & \mathcal{B}\boldsymbol{LP} \\ (\boldsymbol{\pi}-\boldsymbol{u_s})^T & 0 & \boldsymbol{R}^{-1} & \boldsymbol{LP} \\ 0 & (\mathcal{B}\boldsymbol{LP})^T & (\boldsymbol{LP})^T & diag(\tau) \end{pmatrix} \succeq 0 \tag{44b}$$
$$\mathcal{F}_h\boldsymbol{v}_k \le l \cdot \boldsymbol{f}_h \tag{44c}$$
$$\mathcal{G}_h\bar{X}_k \le \boldsymbol{g}_h \tag{44d}$$
$$(29),\ (34),\ (35) \tag{44e}$$

where (44c) and (44d) represent the direct formulation of the constraints (3d) and (9), respectively. The coupling output trajectory transmitted at the current time step k, $\tilde{\boldsymbol{y}}_{z,k} = \mathcal{C}_{zi}\bar{X}_k$. At each sampling time, each subsystem controller solves the optimization problem (44) and the first term of the optimized sequence is applied to the vehicle. Without a strictly guaranteed recursive feasibility, a recovery strategy is implemented as the final defense line. If the original OCP is infeasible for the subsystem Σ_i, the controller loosens the hard constraints (44c) on $\Delta\nu$ and solves the OCP again. The vehicle sacrifices a degree of comfort in exchange for a suboptimal feasible solution.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CONTROLLER PARAMETERS AND MODEL PARAMETERS

$vmax$	maximum allowed velocity	m/s	30
$vmin$	minimum allowed velocity	m/s	0
$smin$	minimum safety distance	m	6.5
$umax$	maximum allowed input	N	1500
$umin$	minimum allowed input	N	-2000
$\Delta umax$	maximum input slew rate	N	250
$\Delta umin$	minimum input slew rate	N	-250
ε_x	level of constraint satisfaction	-	0.05
ε_u	level of constraint satisfaction	-	0.05
τ_m	tolerance constant for message difference	-	0.5
s_c	constant for spacing policy	m	7.5
t_h	time headway for spacing policy	s	0.3
N	predictive horizon length	-	15

V. SIMULATION RESULTS

The platoon consists of 10 vehicles. Table II shows the parameter used in the controllers. Deterministic disturbances range $-40 \leq p^i[1] \leq 350$, $-0.05 \leq p^i[2] \leq 0.05$. Stochastic process disturbances $q^i \sim \mathcal{N}(0, 0.01)$. Stochastic measurement noise $\eta \sim \mathcal{N}(0, 0.01)$. Weighting matrix $Q = diag(1, 0.04)$, $R = 0.1$. The observer is designed as $B_d = C_d = I_{n_x \times n_x}$. The initial state of vehicle i is $(-d_0 * (i-1); 0)$, where $d_0 = 7.5\ m$. The linearization point is $v = 5m/s$, $s = s_c + t_h \times v$, $u = MgR_w C_r \cos\theta + MgR_w \sin\theta$. The values of the model parameters in Table I are: $m = 1722\ kg$, $R_W = 0.315\ m$, $g = 9.8\ N/kg$, $\theta = 0$, $A = 2.6292\ m^2$, $\rho = 1.206\, kg/m^3$, $C_x = 0.2047$, $C_r = 0.0106$, and $T = 0.1\ s$.

To validate the effectiveness of the proposed method, we will evaluate the control performance in terms of the tracking error on velocity, inter-vehicle distance and distance from the leader vehicle. The global distance error is used to refer to the distance tracking error from the leader vehicle to the controlled vehicles, i.e. $(p_o^1 - p_o^i) - (i-1) \times s^{des}$, where p_o is the longitudinal position and $s^{des} = s_c + t_h v^{des}$. Two experiments are presented as follows.

Experiment 1: We compare the result of the proposed method with a robust MPC controller and a normal MPC controller under the PF structure. The tracing velocity is 20 m/s and there exist homogeneous model mismatches and mixed disturbances.

Experiment 2: We compare the result of the proposed method with a PFL strategy when heterogeneous model mismatches and mixed disturbances exist. The desired velocity is 10 m/s.

A. Experiment 1

Experiment 1 aims to demonstrate the advantage of the proposed method in offset cancellation and disturbance attenuation under the same communication condition. Vehicles use the same model and dynamic. Homogeneous model mismatches refer that all the plants update as $A = A + [0\ 0; 0\ 0.01]$.In addition, $p_{[1]}^i = 150$ is a persisting noise; $p_{[2]}^i = 0$ when the longitudinal position $p_o^i \leq 20$, otherwise, $p_{[2]}^i = -0.05$; $q^i \sim \mathcal{N}(0, 0.01)$; $\eta \sim \mathcal{N}(0, 0.01)$. The simulations are performed separately with the proposed method, the robust MPC controller (the constraints are reformulated in the same way and the controller combines the target calculator and the nominal MPC) and the normal MPC controller (using nominal model without considering the disturbances).

The performance is presented with figures of velocity profiles and inter-vehicle distance as shown in Figure 3. Different colored lines in the same picture represent different vehicles. The result shown in Figure 3 indicates that the perturbations are considerably reduced only in the proposed method. The peak magnitudes of the deviation from the equilibrium along the vehicular string are reduced in the proposed method and the robust method. Compared to the normal MPC controller, the overshoot of the last vehicle's speed is reduced by 68.82% and 80.63% in the robust method and the proposed method, respectively; the distance overshoot of the last two vehicles is reduced by 61.28% and 77.83%.

In order to further observe the distribution characteristics of the data at steady state, the box-plots for distance and velocity error are presented in Figure 4 where data points after 20th second are used. On each box, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The averages are plotted using the 'o' symbol.

The ability to suppress random disturbances is reflected in the variance of the process output. In general, all the methods work well on it. The variance of the proposed method is two orders of magnitude higher than the normal method in terms of velocity and one order of magnitude higher in terms of distance. The box plots also show that the results of the robust method and the normal method have more concentrated distributions.

The ability to suppress deterministic disturbances is reflected in the steady-state error of the process output. The average absolute vehicle errors among 10 vehicles $(\frac{1}{10}\sum_{i=1}^{10} |\frac{1}{151}\sum_{k=200}^{350} v_k^i - v_{des}|)$ of the proposed method, the robust method, and the normal method are 0.001 m/s, 0.242 m/s, and 0.432 m/s respectively. Similarly, the average absolute distance errors of the three methods are 0.006 m, 0.105 m, and 0.032 m. Compared to the normal MPC controller, the proposed method reduced the error by 99.75% on velocity and 79.58% on distance.

Overall, the proposed method is an improvement on model predictive control method. The proposed method minimizes the perturbations of not only velocities but also spacing, which indicates that the uncertainties contained in the preceding forecast can be addressed. Under the same communication structure and uncertainty condition, the following behavior of the proposed method shows clear advantages on suppressing overshoot and reducing the offset.

B. Experiment 2

Experiment 2 aims to demonstrate that the proposed method has a comparative advantage to the PFL solution in dealing with dynamics heterogeneity. In this case, the plant i updates as $A = A + [0\ 0; 0\ a^i]$ and $-0.01 < a^i < 0.01$. The parameter a^i among vehicles differs. The assumption of heterogeneity is quite realistic in practice due to the diversity of vehicle models and types and the individual differences in power-train components. The desired speed equals to 10 m/s, and the disturbances are applied the same as experiment 1.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

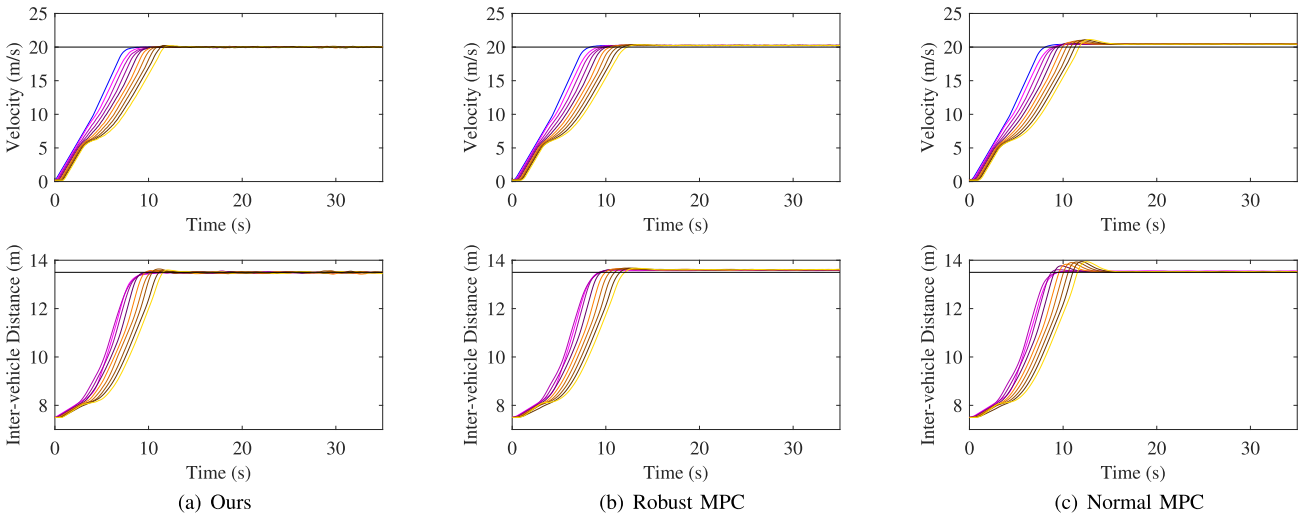


Fig. 3. Comparison result of three methods in experiment 1.

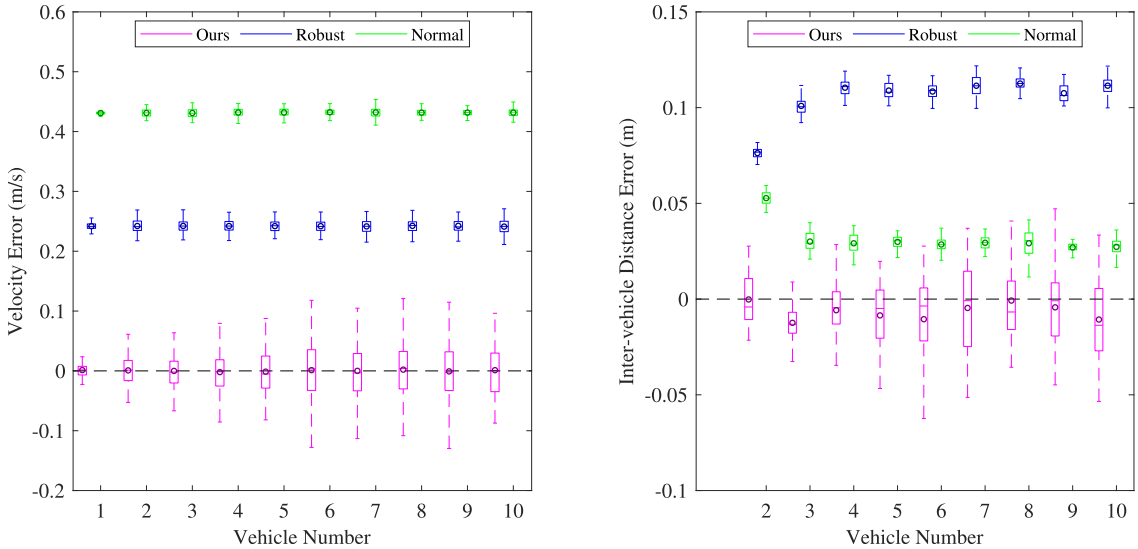


Fig. 4. Boxplots for distance/velocity error in experiment 1.

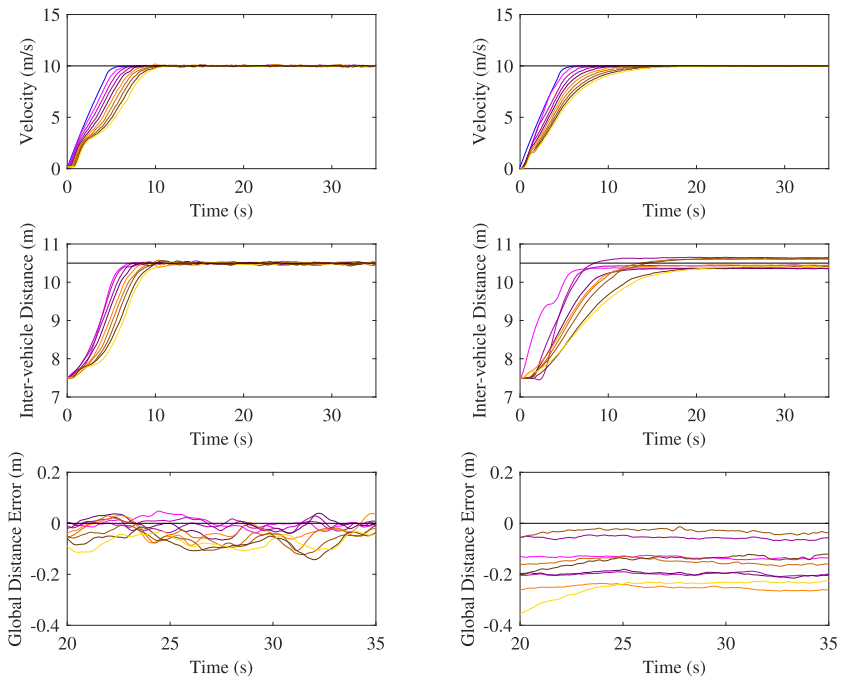
The simulations are performed separately with the proposed method and the PFL strategy. The communication delay between two consecutive vehicles is 0.025s and the delay between the leader and a follower accumulates by vehicle number. The controller sampling interval T=0.1s. Thus, at time k, the leader message received by vehicle 2-5 is sent at time k-T; received by vehicle 6-9 is sent at time k-2T; received by vehicle 10 is sent at time k-3T. In order to make the two strategies more comparable, we adjusted the PFL strategy on the basis of the proposed method. Considering the leader's manipulator means a lot in PFL structure, the leader adopts the proposed method and the followers adopt the robust method as experiment 1 explained. The reason why the following vehicle does not adopt the proposed method is that as the communication delay becomes larger, the uncertainty set becomes larger and renders the problem infeasible.

The performance is shown with figures of vehicle velocity, inter-vehicle distance, and global distance error as Figure 5. The proposed method maintains good performance while the dynamics heterogeneity leads to uneven vehicle spacing and larger steady error of global distance in the platoon under the PFL strategy. Among ten vehicles, the maximum deviation of the mean global distance error ($t \in [20, 35]$) in the proposed method and the PFL structure is 0.081 m and 0.261 m, respectively.

The simulations start from stationary, the time required for the last vehicle in the platoon to reach the desired velocity could represent the mobility of the platoon control strategy. The time spent by the proposed method and the PFL strategy is 11.5s and 22.5s respectively. Communication delay and the PFL topology cause the control performance to be sluggish.

The global distance error should be further discussed considering that the control goal of the PFL strategy is to maintain a corresponding distance from the first car. Under the PFL topology, varying degrees of tracking error on the distance from the leader exists as shown in Figure 6 and the relationship between the global distance error and the model parameter a is approximately linear as represented. The result of the proposed method is marked in red in Figure 6 where the offset exists but is greatly reduced. The slope of the fitted line is close to horizontal, which means the proposed method is almost unaffected by heterogeneous model mismatches.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.



(a) Ours (b) PFL strategy

Fig. 5. Comparison result of two methods in experiment 2.

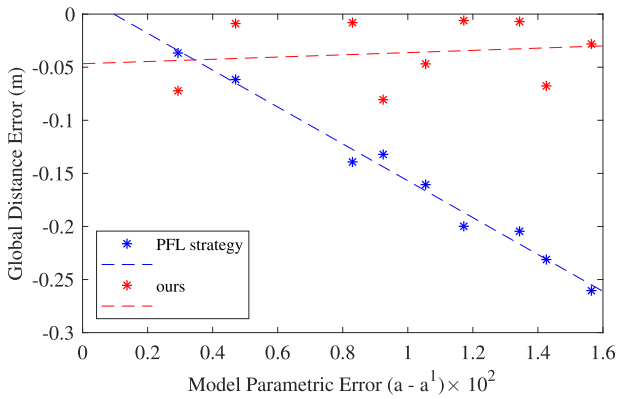


Fig. 6. The relationship between the model parametric uncertainty and the global distance error in experiment 2.

To sum up, the proposed method has a comparative advantage to the PFL solution on mobility and robustness to heterogeneous model mismatch. Shorter response time makes it easier for the platoon to cope with various traffic situations. The ability to handle the heterogeneous model mismatch makes it potential for the new vehicle member to join in. The PF topology has more scalability because it does not need to consider the communication delay that increases with the length of the queue.

VI. Conclusion

This paper proposed a distributed MPC control scheme for vehicle platooning, taking into account the stochastic disturbances, deterministic disturbances, and model uncertainties. The constraints are tightened with respect to the stochastic and deterministic components separately. The target calculator with the observer compensates for the error tending to a steady value, and the MM-MPC rejects the varying disturbance over the predictive horizon.

Existing platoon approaches focus on disturbances or maneuvers of the lead vehicle since generally, the leader faces uncertain traffic situations. Here we suggest a more strict condition where every vehicle in the platoon could be affected by disturbances, in the meantime, only the message of the preceding vehicle can be obtained for the controlled vehicle. Simulation results show that the proposed method is effective in platoon control and achieves offset-free tracking under such conditions. The ability to deal with model uncertainty makes it not only superior to other MPC methods, but also better than the Predecessor Following Leader structure in that it can handle dynamic heterogeneity. The simple Predecessor Following communication brings better mobility and scalability.

The limitations of the current study are summarized as follows. Although the algorithm shows good performance and satisfactory convergence, here we cannot give a theoretical proof of the convergence, and we will work on it in the future. The potential effects of mobility, scalability and practical utility should be considered more carefully in future research, for example, the combination with platoon formation problem is an interesting topic.

References

[1] J. Guanetti, Y. Kim, and F. Borrelli, "Control of connected and automated vehicles: State of the art and future challenges," *Annu. Rev. Control*, vol. 45, pp. 18–40, Apr. 2018.

[2] J. Zhou and H. Peng, "Range policy of adaptive cruise control vehicles for improved flow stability and string stability," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 2, pp. 229–237, Jun. 2005.

[3] S. E. Li *et al.*, "Dynamical modeling and distributed control of connected and automated vehicles: Challenges and opportunities," *IEEE Intell. Transp. Syst. Mag.*, vol. 9, no. 3, pp. 46–58, Jul. 2017.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

[4] Y. Zhou, M. Wang, and S. Ahn, "Distributed model predictive control approach for cooperative car-following with guaranteed local and string stability," *Transp. Res. B, Methodol.*, vol. 128, pp. 69–86, Jan. 2019.

[5] P. Liu, A. Kurt, and U. Ozguner, "Distributed model predictive control for cooperative and flexible vehicle platooning," *IEEE Trans. Control Syst. Technol.*, vol. 27, no. 3, pp. 1115–1128, May 2019.

[6] S. Zhao and K. Zhang, "A distributionally robust stochastic optimization-based model predictive control with distributionally robust chance constraints for cooperative adaptive cruise control under uncertain traffic conditions," *Transp. Res. B, Methodol.*, vol. 138, pp. 144–178, Jun. 2020.

[7] N. Chen, M. Wang, T. Alkim, and B. Van Arem, "A robust longitudinal control strategy of platoons under model uncertainties and time delays," *J. Adv. Transp.*, vol. 2018, Apr. 2018, Art. no. 9852721.

[8] Y. Zheng, S. E. Li, K. Li, F. Borrelli, and J. K. Hedrick, "Distributed model predictive control for heterogeneous vehicle platoons under unidirectional topologies," *IEEE Trans. Control Syst. Technol.*, vol. 25, no. 3, pp. 899–910, Dec. 2017.

[9] Y. J. Wang and J. B. Rawlings, "A new robust model predictive control method I: Theory and computation," *J. Process Control*, vol. 14, no. 3, pp. 231–247, Apr. 2004.

[10] G. Na, G. Park, V. Turri, K. H. Johansson, H. Shim, and Y. Eun, "Disturbance observer approach for fuel-efficient heavy-duty vehicle platooning," *Veh. Syst. Dyn.*, vol. 58, no. 5, pp. 1–20, 2019.

[11] L. Tsuei and Ö. Savaü, "Transient aerodynamics of vehicle platoons during in-line oscillations," *J. Wind Eng. Ind. Aerodyn.*, vol. 89, no. 13, pp. 1085–1111, Oct. 2001.

[12] M. Sybis, M. Rodziewicz, and K. Wesolowski, "Influence of sensor inaccuracies and acceleration limits on IEEE 802.11p-based CACC controlled platoons," in *Proc. IEEE 91st Veh. Technol. Conf. (VTC-Spring)*, May 2020, pp. 1–6.

[13] N. Sockeel, J. Shi, M. Shahverdi, and M. Mazzola, "Sensitivity analysis of the vehicle model mass for model predictive control based power management system of a plug-in hybrid electric vehicle," in *Proc. Transp. Electrific. Conf. Expo.*, New York, NY, USA, Jun. 2018, pp. 767–772.

[14] A. Mesbah, "Stochastic model predictive control an overview and perspectives for future research," *IEEE Control Syst. Mag.*, vol. 36, no. 6, pp. 30–44, Feb. 2016.

[15] X. Xu, J. Zhang, and L. Tang, "String exponential stability with mode constraint of stochastic vehicle following systems," *J. Dyn. Syst., Meas., Control*, vol. 135, no. 6, Nov. 2013, Art. no. 061011.

[16] C. Somarakis, Y. Ghaedsharaf, and N. Motee, "Risk of collision in a vehicle platoon in presence of communication time delay and exogenous stochastic disturbance," in *Proc. IEEE Conf. Decis. Control*, New York, NY, USA, Dec. 2018, pp. 4487–4492.

[17] Y. Zhou, S. Ahn, M. Chitturi, and D. A. Noyce, "Rolling horizon stochastic optimal control strategy for ACC and CACC under uncertainty," *Transp. Res. C, Emerg. Technol.*, vol. 83, pp. 61–76, Oct. 2017.

[18] F. Gao, X. Hu, S. E. Li, K. Li, and Q. Sun, "Distributed adaptive sliding mode control of vehicular platoon with uncertain interaction topology," *IEEE Trans. Ind. Electron.*, vol. 65, no. 8, pp. 6352–6361, Aug. 2018.

[19] Y. Zhu and F. Zhu, "Barrier-function-based distributed adaptive control of nonlinear CAVs with parametric uncertainty and full-state constraint," *Transp. Res. C, Emerg. Technol.*, vol. 104, pp. 249–264, 2019.

[20] A. Petrillo, A. Salvi, S. Santini, and A. S. Valente, "Adaptive multi-agents synchronization for collaborative driving of autonomous vehicles with multiple communication delays," *Transp. Res. C, Emerg. Technol.*, vol. 86, pp. 372–392, Jan. 2018.

[21] J. Ploeg, E. Semsar-Kazerooni, G. Lijster, N. Van De Wouw, and H. Nijmeijer, "Graceful degradation of cooperative adaptive cruise control," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 1, pp. 488–497, Feb. 2015.

[22] Z. Wang, Y. Bian, S. E. Shladover, G. Wu, S. E. Li, and M. J. Barth, "A survey on cooperative longitudinal motion control of multiple connected and automated vehicles," *IEEE Intell. Transp. Syst. Mag.*, vol. 12, no. 1, pp. 4–24, Dec. 2019.

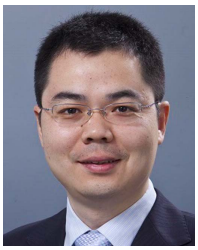
[23] C. Mark and S. Liu, "Distributed stochastic model predictive control for dynamically coupled linear systems using probabilistic reachable sets," in *Proc. 18th Eur. Control Conf. (ECC)*, Jun. 2019, pp. 1362–1367.

[24] M. Farina, L. Giulioni, and R. Scattolini, "Distributed predictive control of stochastic linear systems with chance constraints," in *Proc. Amer. Control Conf. (ACC)*, Jul. 2016, pp. 20–25.

[25] L. Xie, H. Su, S. Lu, and L. Xie, "Linear mismatched model based offset-free MPC for nonlinear constrained systems with both stochastic and deterministic disturbances and its application to CSTR," *IEEE ACCESS*, vol. 6, pp. 69172–69184, 2018.

Xiaorong Hu received the B.S. degree from Zhejiang University, Hangzhou, China, in 2018, where she is currently pursuing the Ph.D. degree with the College of Control Science and Engineering. Her current research interests include platoon control and model predictive control.

Lantao Xie received the Ph.D. degree from Zhejiang University in 2019. He is currently an Algorithm Engineer with Huawei Technologies Company Ltd. His research interests include model predictive control and machine learning.



Lei Xie received the B.S. and Ph.D. degrees from Zhejiang University, China, in 2000 and 2005, respectively. From 2005 to 2006, he was a Post-Doctoral Researcher at the Berlin University of Technology, and an Assistant Professor from 2005 to 2008. He is currently a Professor at the Department of Control Science and Engineering, Zhejiang University. To date, his research activities culminated in over 30 papers that are published in internationally renowned journals and conferences, three book chapters, and a book in the area of applied multivariate statistics and modeling. His research interests include the interdisciplinary area of statistics and system control theory.

Shan Lu received the B.S. and Ph.D. degrees from Zhejiang University, China, in 2011 and 2016, respectively. He is currently an Assistant Professor with the Institute of Intelligence Science and Engineering, Shenzhen Polytechnic. His research interests include modeling and optimization of supply chain scheduling, machine learning, and process modeling.

Weihua Xu was born in 1976. She received the Ph.D. degree in control science and engineering from Zhejiang University in 2003. From 2003 to 2005, she was an Assistant Professor with the Institute of Advanced Process Control, Zhejiang University. Since 2006, she has been an Associate Professor at the Institute of Cyber-Systems and Control, Zhejiang University. Her current research interest includes processing control.

Hongye Su was born in 1969. He received the B.S. degree in industrial automation from the Nanjing University of Chemical Technology, Jiangsu, China, in 1990, and the M.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1993 and 1995, respectively. He was a Lecturer with the Department of Chemical Engineering, Zhejiang University, from 1995 to 1997, where he was an Associate Professor with the Institute of Advanced Process Control from 1998 to 2000, where he is currently a Professor with the Institute of Cyber-Systems and Control. His current research interests include the robust control, time-delay systems, and advanced process control theory and applications.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.